

Brennan Mahoney

bmm1@bu.edu | (774) 280-0845 | [Portfolio Website](#) | <https://www.linkedin.com/in/brennan-mahoney> | <https://github.com/bmahoney1>

Education

Northbridge High School

Awards: Valedictorian, Harvard Book Award Recipient

Boston University College of Engineering

May 2025

Bachelor of Science in Computer Engineering and Concentration in Machine Learning

GPA: 3.85/4.00 • Latin Honors • Dean's List (7x)

Skills

Languages & Frameworks: Python, JavaScript, C++, Java, SQL, PyTorch, LangChain, React, React Native, Node.js, FastAPI

Cloud & DevOps: Docker, Azure ML, Databricks, Jenkins, GitLab, Terraform, Kubernetes, AWS (EC2, S3, Cognito, DynamoDB), New Relic, Git

Machine Learning Domains: Generative Models, Computer Vision, Deep Learning, Data Engineering, Reinforcement Learning, MLOps, NLP

Work Experience

Full-Stack Software Development Engineer

October 2025 – Present

CVS Health

- Named co-inventor on a filed patent application for an AI remediation agent to detect, analyze, and automatically patch Snyk vulnerabilities across 50+ GitHub repositories, reducing manual engineering remediation effort by 75%.
- Created FastAPI service that extracts store-hour changes, publishes messages to Kafka, and is used by Airflow jobs to enable customer outreach.
- Designed scalable connectors to orchestrate data transfer pipelines across Snowflake, Kafka, and MongoDB for enterprise data integration.
- Developed packages for MongoDB, Kafka, and Redis connection across repositories, improving consistency and reducing integration overhead.

Associate Full-Stack Software / Data Engineer

May 2025 – October 2025

John Hancock

- Designed and deployed machine learning pipelines, leveraging Databricks for model training with enterprise datasets.
- Built and queried vector databases to perform semantic similarity analysis on call transcripts using cosine similarity, identifying sensitive personal information and validating question relevance for compliance and accuracy.
- Developed data ingestion pipelines using Azure DF and Databricks, automating ETL workflows for structured and unstructured financial data.

Full-Stack Software Engineering Intern → Contractor

May 2024 – May 2025

John Hancock

- Implemented and deployed geofencing functionality that prompted users to enable location services and captured timestamps when users were over 1,500 meters from care addresses, helping reduce fraudulent claims through improved location verification.
- Created a monitoring system using New Relic, allowing the team to detect and respond to issues proactively, improving app reliability.
- Built new backend queries between CareGiver and Salesforce Data Cloud for legacy claims, enhancing access to patient information.

Graduate Level Deep Learning Teaching Assistant

January 2025 – May 2025

Boston University

- Supported instruction of advanced deep learning topics including convolutional neural networks (CNNs), recurrent neural networks (RNNs), transformers, generative models (VAEs, GANs), and optimization techniques (SGD, Adam, dropout, batch norm).
- Graded and debugged PyTorch-based homework and projects, with tasks such as classification and sequence/generative modeling.
- Provided technical support on deep learning workflows through office hours, helping students implement, train, and evaluate neural network models using the Shared Computing Cluster (SCC) and GPU resources.

Full-Stack Software Engineer

September 2024 – February 2025

RedStone (Boston University Spark!)

- Developed frontend using React for an intuitive UI and implemented database schemas to support efficient data storage and retrieval.

Projects/Extra Curricular

AI Personal Assistant (Company Gen AI Hackathon Winner)

- Built an AI personal assistant that triages unread emails and automates follow-ups, meeting scheduling, and task creation within Google Workspace (Gmail, Calendar, Tasks).
- Engineered an agentic LangGraph workflow with specialized summarization, reasoning, and action agents.
- Implemented human-in-the-loop approval, persistent state/session management, and Google API integrations to execute approved actions while maintaining thread context and preventing duplicate operations.

RoomformAI (Team Leader)

- Developed a web app enabling users to design/visualize rooms in 3D, using React, Three.js, Firebase, and AWS; selected for YC interview.
- Implemented a 2D-to-3D furniture model generation pipeline using diffusion-based models (Bi-RefNet, Trellis), hosted on AWS EC2 with PyTorch (CUDA), for realistic 3D outputs from 2D images.
- Implemented LiDAR and an OpenAI-powered AI agent using LangGraph, enabling contextual room analysis and automation.
- Integrated NLP techniques to parse user queries and stored Pinecone embeddings for semantic furniture search.

Audio Style Transfer via Neural Network (Team Leader)

- Collaborated on a team to implement a Convolutional Neural Network (CNN) setup with a VGG19 network for style transfer in audio spectrograms, bridging the gap between image processing and audio data.
- Employed machine learning models for audio synthesis, specifically using the WaveNet vocoder, to convert mel spectrograms back to audio, demonstrating proficiency in both audio analysis and synthesis.

Pi Kappa Alpha Fraternity (Lead Recruitment Chairman)

- Oversaw recruitment efforts by organizing and facilitating meetings to identify new members who aligned with the fraternity's values.
- Led and participated in philanthropy fundraisers, helping raise over \$100,000 for inclusive foundations.